

Data Analysis Final Assignment Report

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1 Contributions

- Kornel Lengl:
 - Data quality analysis and preprocessing pipeline
 - Visualizations and EDA
 - Probability analysis tasks
 - Report and presentation making
- Juanith Mathew Thomas:
 - Dataset selection and acquisition
 - Regular and polynomial-2 linear regression
 - PCA and t-SNE analysis
 - Report and presentation making

2 Dataset Description

The chosen data set is ***Rossmann Store Sales*** from kaggle.com[1]. This data set is a collection of daily sales of 1115 Rossman stores over a period of more than two years.

The data set contains daily observations for each store, with a *Date* column acting as a chronological index for every store day record. This gives the data a regular temporal structure, which makes it especially suitable for time-series analysis.

The covered period of time starts with 01.01.2013 and ends 31.07.2015 with consecutive days.

The key daily variables are

- Store - a unique Id for each store
- Date - the day when the data is originated
- DayOfWeek - weekly pattern indicator
- Sales - the turnover for any given day (in euro)
- Customers - the number of customers on a given day
- Open - an indicator for whether the store was open: 0 = closed, 1 = open
- StateHoliday - indicates a state holiday

- SchoolHoliday - indicates the closure of public schools
- Promo - indicates whether a store is running a promo on that day

Size and structure:

- Dataset shape: 1,017,209 x 9 (rows x columns)
- 750-980 rows for each store

The data were originally well sanitized; there are *no gaps* or *duplicates* in the data set. There were *no missing values* identified in the data set.

3 Task 1. Data Preprocessing and Basic Analysis

3.1 Basic statistical analysis using pandas

406,974 data rows were extracted for this study; these data correspond to one year from 01/01/2013 to 31/12/2013.

	Store	DayOfWeek	Sales	Customers	Open	Promo	SchoolHoliday
count	406974.000000	406974.000000	406974.000000	406974.000000	406974.000000	406974.000000	406974.000000
mean	557.998943	3.994525	5658.533675	629.043686	0.830380	0.369864	0.188774
std	321.872731	1.999995	3821.166174	464.764530	0.375299	0.482768	0.391329
min	1.000000	1.000000	0.000000	0.000000	0.000000	0.000000	0.000000
25%	279.000000	2.000000	3615.000000	394.000000	1.000000	0.000000	0.000000
50%	558.000000	4.000000	5598.000000	603.000000	1.000000	0.000000	0.000000
75%	837.000000	6.000000	7691.000000	833.000000	1.000000	1.000000	0.000000
max	1115.000000	7.000000	38025.000000	7388.000000	1.000000	1.000000	1.000000

Figure 1: Descriptive stats for key variables

3.2 Original data quality analysis and data preprocessing

11,167 outlier store entries (2.74%) were identified but not removed. Sales spikes can be caused by promotions or holidays; removing them would erase genuine and impactful events rather than errors. To identify the outliers the 25th and 75th percentile were used.

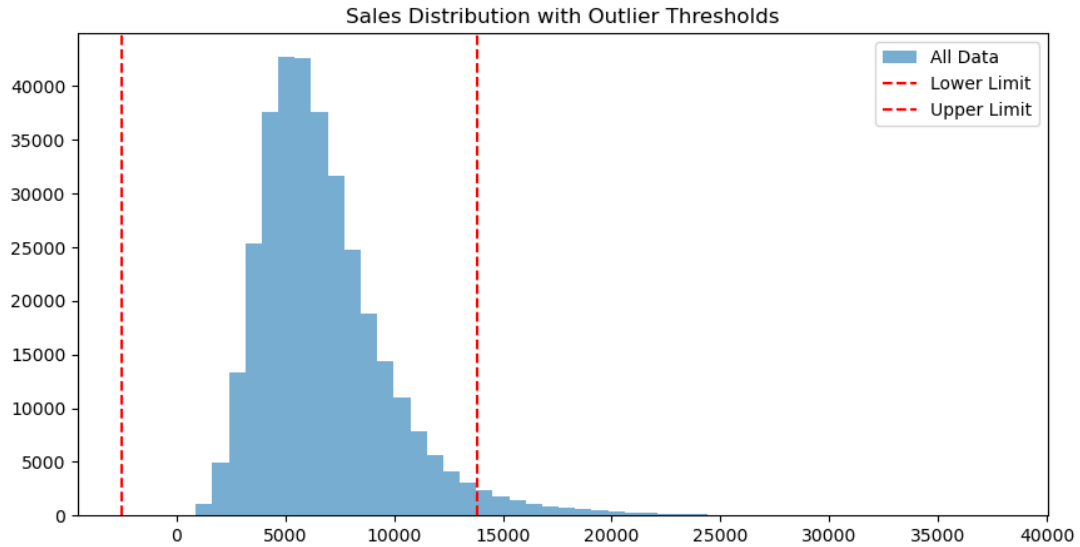


Figure 2: Outlier identification

69,050 (16.97%) data entries were removed, because of zero sales. These entries do not indicate errors in the data set, but a closed sales day.

Duplicates were checked for, none present.

Some values had to be **feature engineered**. The feature *IsWeekend* was added. It is a binary value, which indicates if a sales day is Saturday or Sunday.

The Date of data type `datetime64` was broken into 2 features namely *date* and *month*. Only *date* and *month* can be used for linear regression because of it's numerical data type.

3.3 Preprocessed vs original data visual analysis

Firstly, the size of the dataset reduced. Instead of having 700-1000 entries for each shop, skewed in favor of some, the data was sieved to obtain 365 entries of each 1115 shop. Days on which stores had 0 sales were removed since they could negatively impact the pattern learning of the LR model.

The data engineered value set *IsWeekend* might help the linear regression better predict the sales of a shop that remains open (and has more sales) on the weekend.

Original dataframe: 1,017,209 rows $\times$ 9 columns

Preprocessed dataframe: 337,924 rows $\times$ 12 columns

4 Task 2. Visualization and Exploratory Analysis

4.1 Time series visualizations

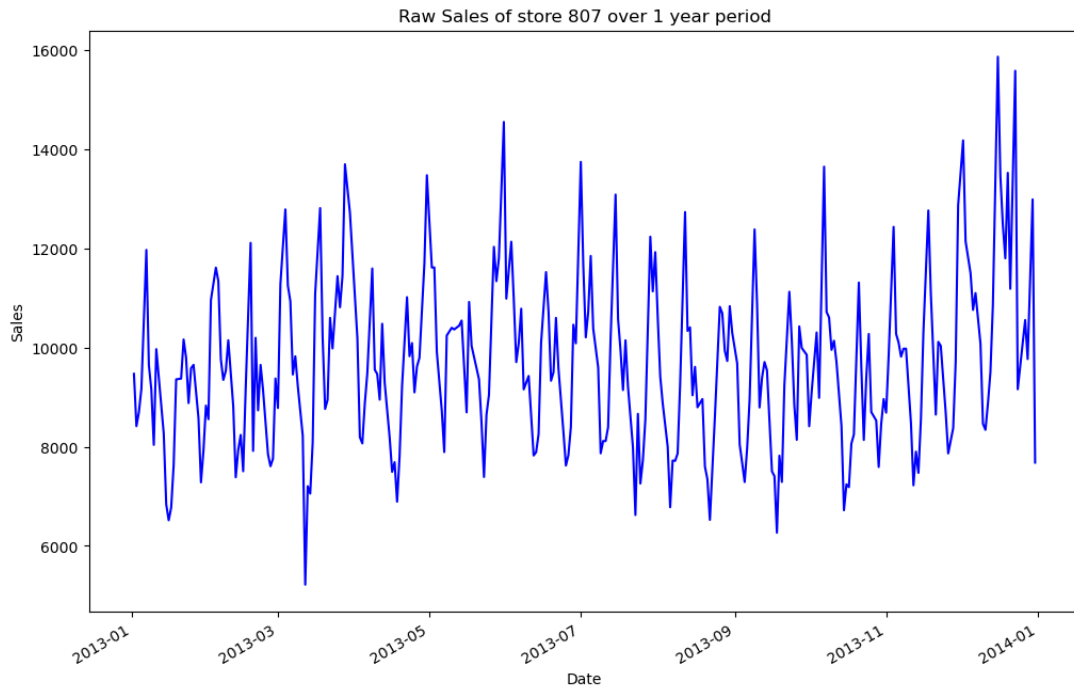


Figure 3: Raw sales data for a single store

Figure 3 shows that the daily sales of the individual stores vary heavily, with the **lowest sales during March** and the **highest sales during December**.

4.2 Distribution analysis with histograms

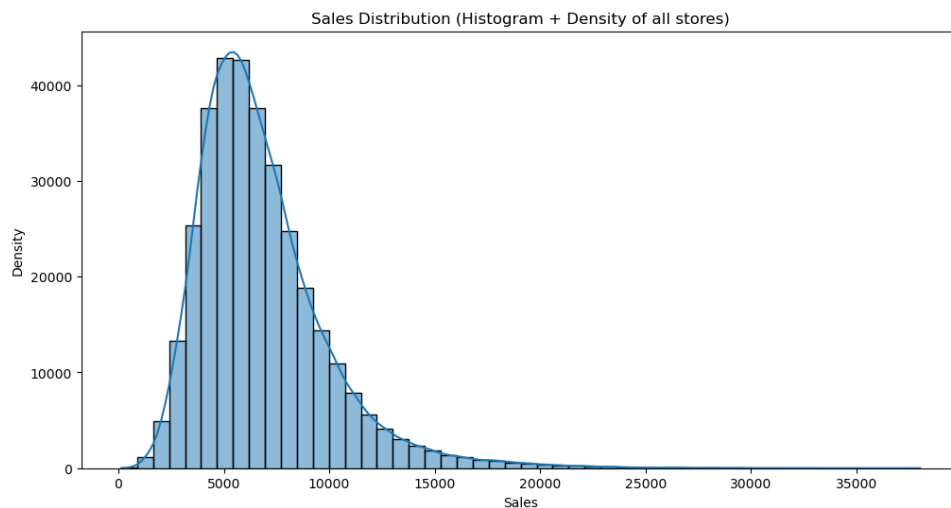


Figure 4: Histogram of daily sales of all 1115 stores

The histogram shows a slight right skewness. This is caused by the outliers that were noted but not removed.

4.3 Correlation analysis and heatmaps

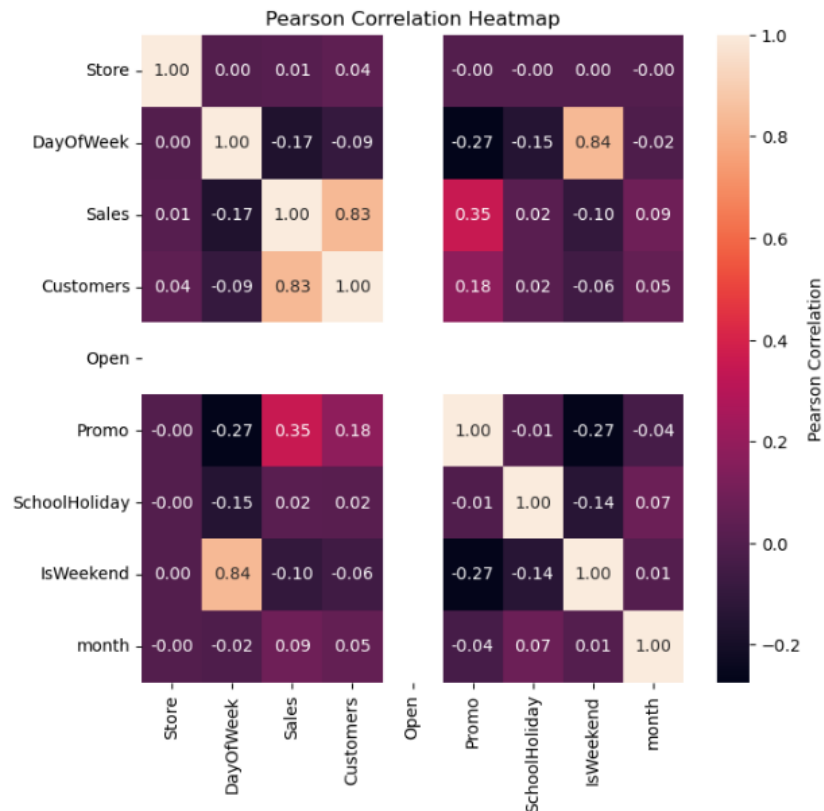


Figure 5: Pearson or Spearman heatmap

The heatmap shows expected correlations, for example the correlation between the number of *customers* and *sales*. It also shows unexpected lack of correlation, like between *promo days* and *sales*.

The correlation is quite similar for both heatmaps in most aspects, subtle differences caused by the difference in method of correlation calculation.

Pearson correlation is the better method since the dataset:

- Doesn't have significant outliers
- Not heavily skewed
- Is sufficiently large

4.4 Daily pattern analysis

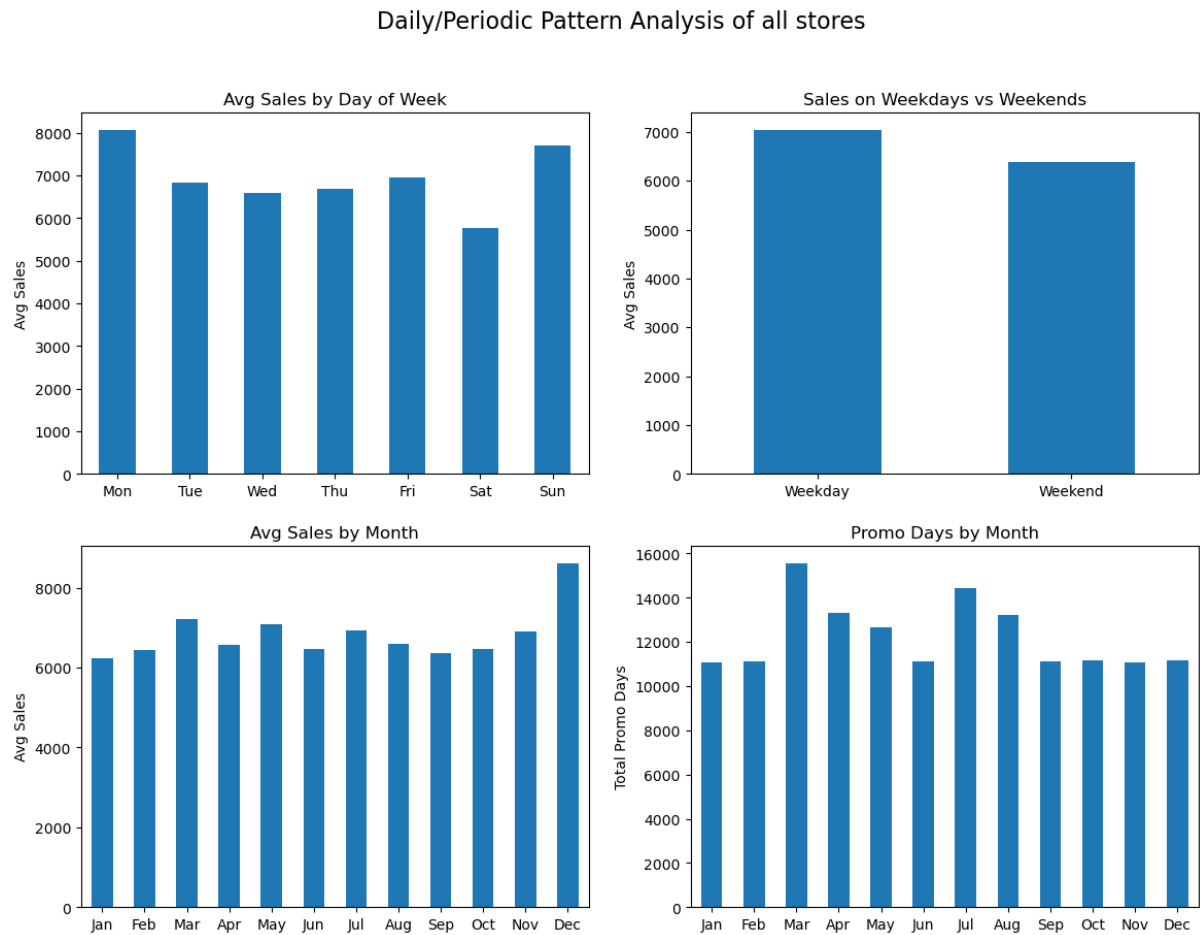


Figure 6: Day of the week and monthly patterns

4.5 Summary of observed patterns

- Although individual stores show large deviations in sales across different days of the week, across all 1,115 stores Monday is the strongest sales day.
Evidence: True, in the Average Sales by Day of Week bar plot, Mondays have the highest mean sales, and Saturdays the lowest.
- Weekdays always have higher sales days than weekends.
Evidence: True, the sales on Weekdays vs Weekends plot shows a clear preference for weekdays.
- March is the strongest sales since it has most Promos.
Evidence: False, the Average Sales by Month (Seasonality) plot clearly shows that most sales occur in December.
- Seasonality has a stronger influence on sales than promotions.
Evidence: True, the Promo Days by Month plot shows that December has the fewest promo days, yet the Average Sales by Month (Seasonality) plot shows it has the highest sales. March has the most promo days and the second-highest sales, which indicates that although promotions have less impact than seasonality, they are still effective.

5 Task 3. Probability Analysis

5.1 Threshold-based probability estimation

The chosen threshold event is "high sales days", which are arbitrary defined as the 90th percentile of sales days in turns of daily turnover.

This threshold clearly indicates an extraordinary event that is worth of analyzing.

For threshold-based probability estimation it makes sense to consider individual stores instead of the average of all the stores, simply because a sales day that is extraordinary for one specific store is might not special for another store.

Lets consider store number 337:

- High-sales threshold (90th percentile): 6242.6 Euros
- High-sales event count: 31 from 303
- Empirical probability $P(\text{event})$: 10.231%

5.2 Cross tabulation analysis

The following figure shows the cross tabulation of two features. It clearly shows that promotion days are much less likely during the weekend than they are on weekdays.

IsWeekend	0	1
Promo		
0	55.29	44.71
1	80.30	19.70

Figure 7: Cross tabulation of *Promo* and *IsWeekend*

5.3 Conditional probability analysis

For conditional probability analysis, the probability of the *High Sales* days (above 90th percentile) and *Low Sales* days (below 10th percentile) were analyzed with the condition of Promotion day. Continuing the example of store number 337:

- $P(\text{HighSales} | \text{Promo} = 1)$: 17.42 %
- $P(\text{HighSales} | \text{Promo} = 0)$: 4.68 %
- $P(\text{LowSales} | \text{Promo} = 1)$: 2.27 %
- $P(\text{LowSales} | \text{Promo} = 0)$: 16.37 %

5.4 Summary of observations from each probability task

- Promo days show a much higher chance of *High Sales* than non-promo days, so promotions are strongly associated with extreme high sales in this dataset.
- Consecutively, on days where there are no promos, the chance of *Low Sales* increases significantly.
- Promo isn't assigned randomly, stores may run promos on special dates, high-demand periods, or in specific store types, so figures can be inflated by other factors (season, holidays, store location/type).

6 Task 4. Statistical Theory Applications

6.1 Law of Large Numbers (LLN) demonstration

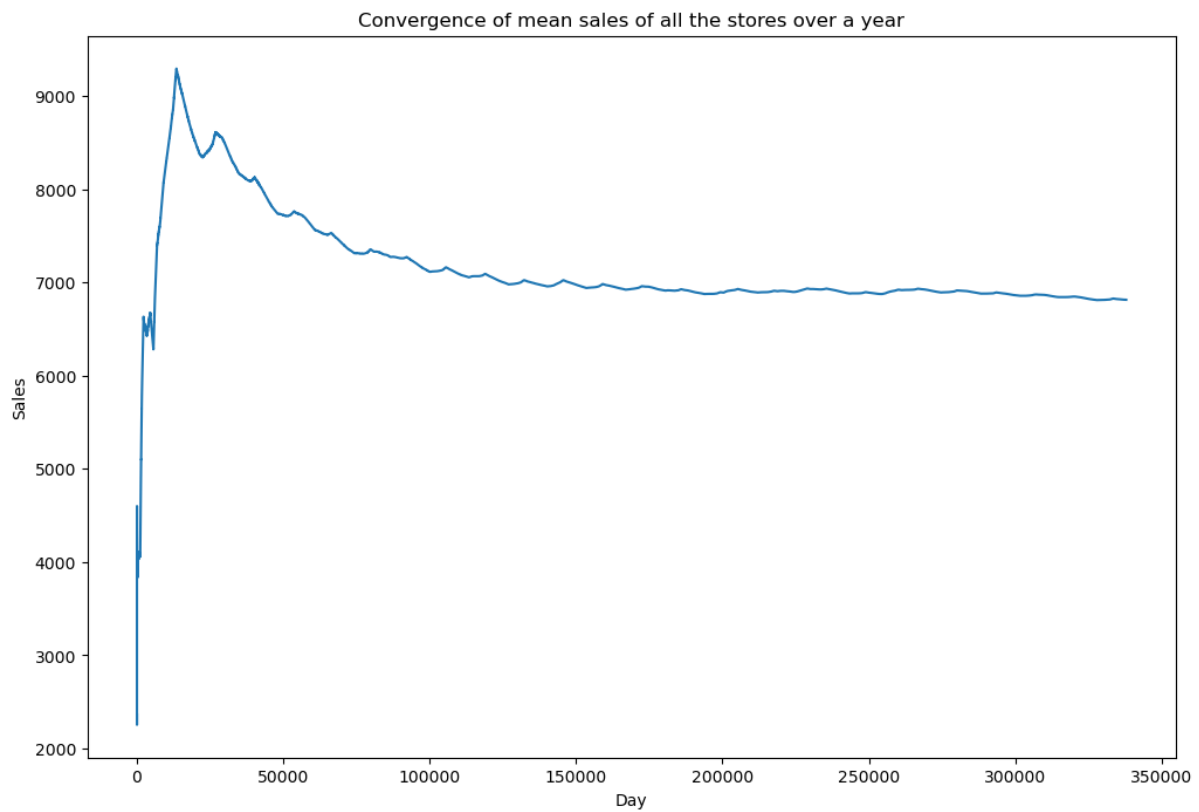


Figure 8: Law of large numbers

Figure 8 shows how the mean of sales of all stores converge over one year worth of sales days.

6.2 Central Limit Theorem (CLT) application

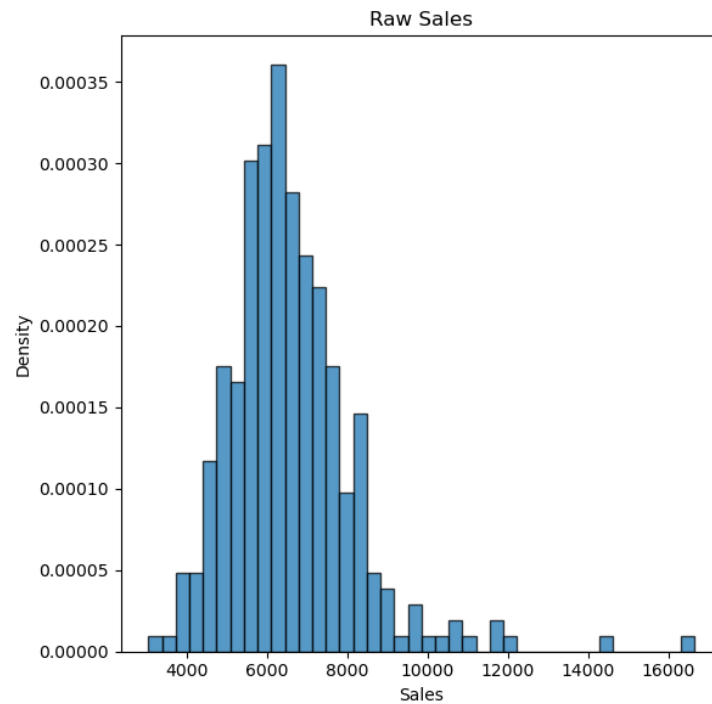


Figure 9: Central Limit Theorem applied with $n=1$

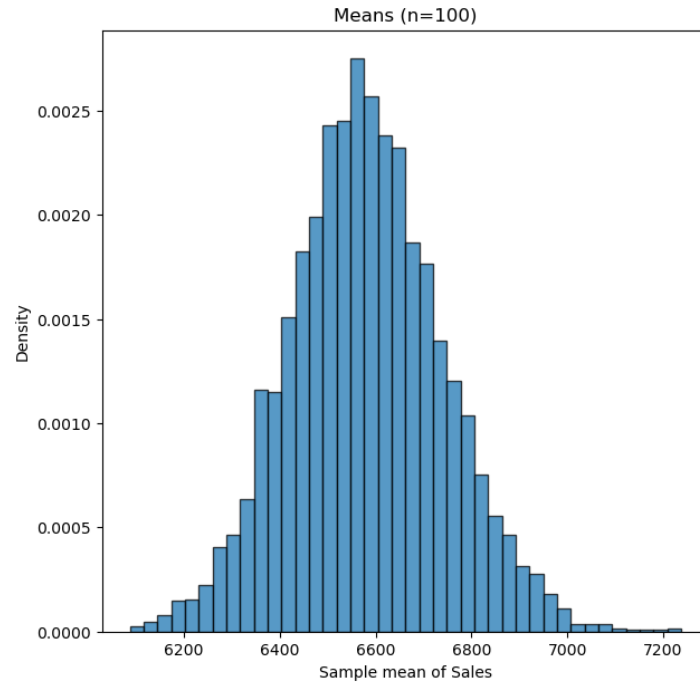


Figure 10: Central Limit Theorem applied with $n=100$

6.3 Result interpretation

- The running mean of Sales moved a lot early on and then stabilized near the overall average as more days were included, which is the qualitative behavior LLN predicts.
- CLT showed that larger n produced a more even bell-shaped distribution of means and a narrower spread.

7 Task 5. Regression Analysis

7.1 Linear or Polynomial model selection

- The target variable $[y]$ is Sales. The feature matrix $[X]$ contains all remaining predictors.
- Motivation: Two candidate models were tested: (1) a standard linear regression model as a simple, interpretable baseline, and (2) a degree-2 polynomial expansion followed by linear regression to allow mild non-linear effects and interactions. The polynomial model was included because the relationship between predictors and sales may not be strictly linear, and a more flexible curve can sometimes capture patterns that a straight line cannot. Although the results made it clear that there was no significant difference.
- The dataset contains daily records across many stores and a large number of observations (about 337,000 rows). A randomized 75/25 train-test split was used to ensure both sets include a broad encapsulation of stores and time periods, while still reserving a substantial subset for evaluation.

7.2 Model fitting and validation

- After performing the train-test split, one-hot encoding was used to convert categorical features into numeric format that the model can use. Model fitting was performed using a scikit-learn Pipeline so preprocessing and training happen consistently in one workflow.
- A holdout method was used for validation where 25% of the total dataset was used for testing. A fixed `random_state` was set to make the split reproducible across runs.
- Multiple metrics were computed, represented here:

Metric	Simple	Polynomial (deg=2)
RMSE	1560.96	1477.21
R^2	0.75	0.78
MAE	1114.07	1066.37

Table 1: Test-set performance comparison of the two regression models.

- Some important things to note are:
 - The polynomial-2 + linear regression model resulted in a Mean Absolute Percentage Error (MAPE) of 16.89%. This metric states that the model is able to predict sale patterns with a decent accuracy but not ideal yet.
 - External notebooks referenced employ methods like XGBoost and Random Forest. These models tend to be more accurate when predicting the sales returning a MAPE as low as 9.6%.

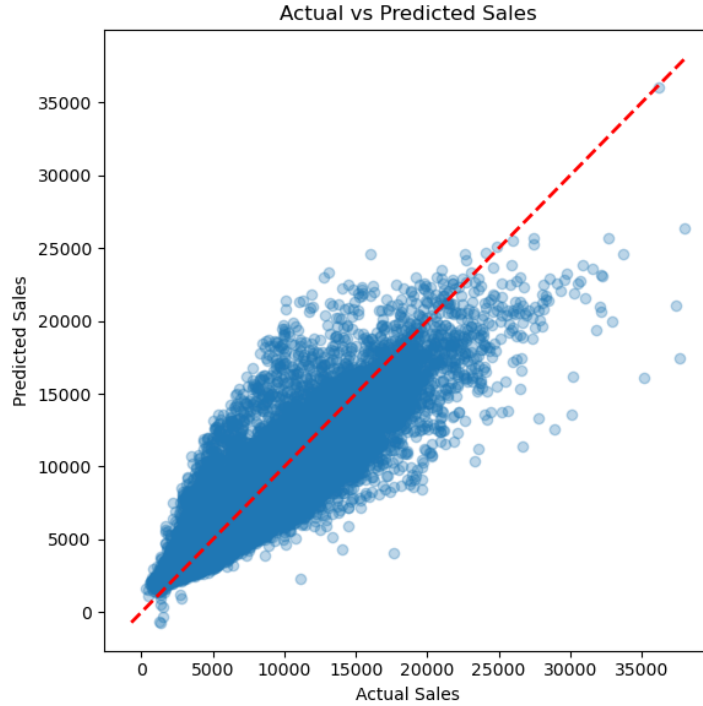


Figure 11: Reported sales vs. Predicted sales for Polynomial-2 + linear regression

7.3 Result interpretation and analysis

- Main effects and practical meaning: With the **degree-2 polynomial + linear regression model**, Sales is explained by the usual “direct” effects of each feature.

However, the effect of a feature isn’t always a fixed amount. For example, changing a feature might increase sales more at some values than others, and the impact of one feature can change depending on another feature.

Because this polynomial model gave **slightly better** $RMSE/MAE$ and R^2 than the plain linear model, it suggests the Sales patterns in the data are not perfectly straight-line, and adding a bit of curvature/interactions helps the model fit better.

- Failure cases or where model performs poorly: Retaining the outliers does help in better model tuning. It comes with it’s own set of disadvantages like pulling least-squares models strongly and resulting in distortion of the fit. The data volume is high and the random splits can leak future patterns into training and make results look better than real deployment.

8 Key Findings and Conclusions

- Main findings from preprocessing and EDA: The dataset is mostly complete, no duplicates were found, no missing values. Rows with 0 values were present for days where the stores were not open, or it was a holiday, etc. Outliers were present but not many, none were removed.
- Main findings from probability tasks: Most important observation was that the HighSales are strongly associated with Promo and when Promo is not running, LowSale days shoot up.
- Main findings from LLN and CLT: The law of large numbers becomes apparent not when just using the whole dataset but when looking at a random individual store, the convergence of mean can be seen, although creeping up due to December sales being higher over the board.

- Main findings from regression: The prediction itself is quite complex for this dataset. The polynomial 2 model can perform predictions with a decent accuracy, displaying an understanding of store and customer patterns. t-SNE shows distinct islands conveying strong distinction between groups of rows that have different underlying feature patterns (i.e., observations that are similar cluster together, and dissimilar ones form separate clusters).
- Limitations: Outlier characteristics may appear in the test set, but predictions may still miss extreme sales values because the model is fitted to capture the overall average trend across the full dataset rather than rare, unusual cases.
- What you would do next if you had more time: Firstly, the learning graph can be analyzed to see training score and validation/test score. Overfitting/Underfitting etc. can also be checked.
In case of more time, adapting the notebook to a simpler time-series dataset with linear regression could result in better demonstration of more accurate predictions.

9 Reproducibility Notes

- Exact dataset source link and version or download date:
<https://www.kaggle.com/datasets/pratyushakar/rossmann-store-sales/data> (Version 3)
- Key libraries used and versions (optional but recommended):
 - python: 3.13.5 — packaged by Anaconda, Inc. — (main, Jun 12 2025, 16:37:03) [MSC v.1929 64 bit (AMD64)]
 - numpy: 2.1.3
 - pandas: 2.2.3
 - scikit-learn/sklearn: 1.6.1
 - scipy: 1.15.3
 - matplotlib: 3.10.0
 - seaborn: 0.13.2
 - umap-learn: 0.5.11
 - random: 3.13.5
- How to run the notebook end-to-end: No special instructions need to be used. As long as the whole repository is checked out without discrepancies, all resources for running the notebook will be automatically loaded.

References

- [1] “Rossmann store sales,” Kaggle, 2019, [Accessed on 20.01.2026]. [Online]. Available: <https://www.kaggle.com/datasets/pratyushakar/rossmann-store-sales/data?select=store.csv>